Introduction to Financial Markets

Arun Kumar

Department of Mathematics, IIT Ropar

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Outline

1. Financial Markets
2. Bonds Market
3. Stock Market
4. Stock Market Indexes
5. FX Market
6. Financial Derivatives
7. Important Factors for Cross Sectional Study
8. Technical Analysis
9. References

Financial Markets

- Financial markets match buyers and sellers to set a price for financial assets.
- Financial markets help to bring people together so money flows to where it is needed most.
- Financial markets provide an opportunity to invest money to build up money for the future.
- It allows people to take out life, non-life, medical insurances.
- It enable banks to borrow money, helping them to make loans to people wishing to borrow.

The following are the major asset classes.

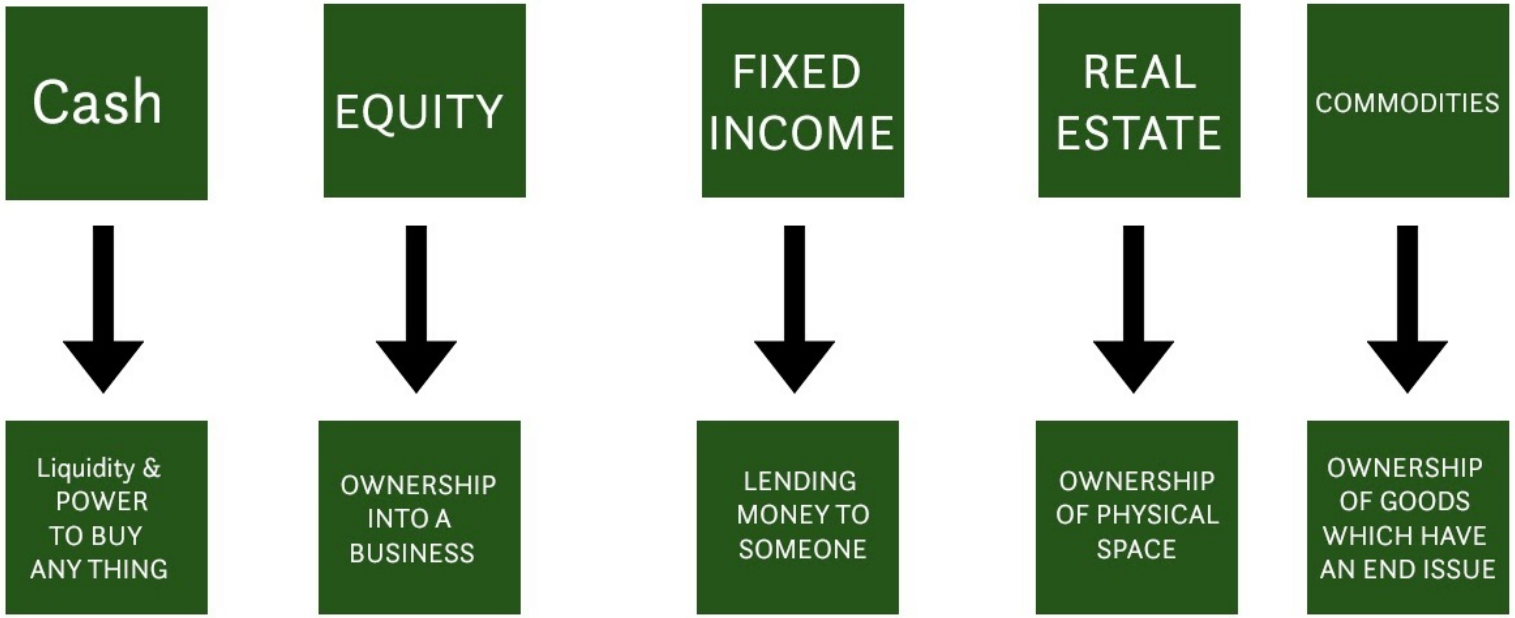


Figure: Major Asset Classes

Definition

Financial instruments are assets that can be traded, or they can also be seen as packages of capital that may be traded.

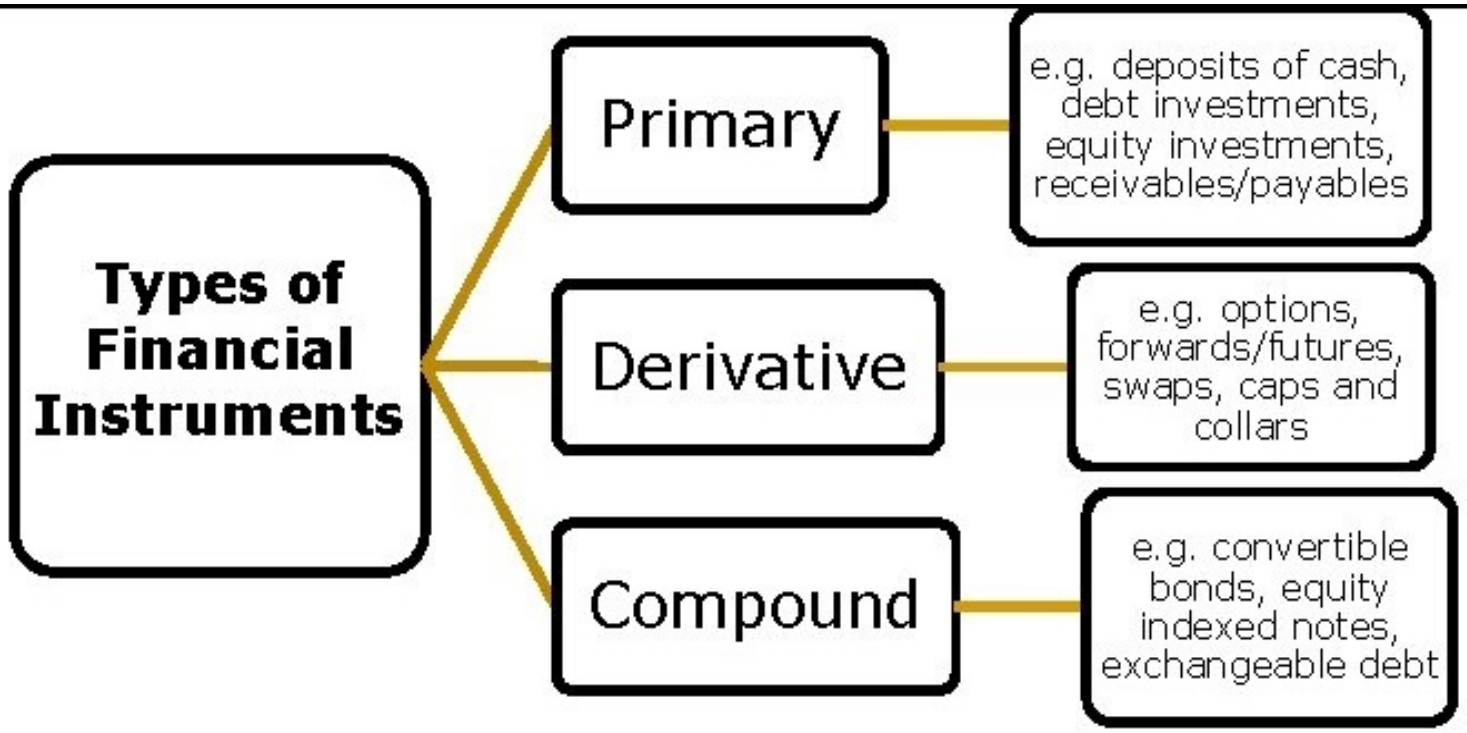


Figure: Financial Instruments

Typical Job Profiles in Financial Fields

- Portfolio Manager
- Corporate Finance Manager
- Accounts Manager
- Risk Manager
- Investment Banker
- Traders
- Economist
- Quant
- Financial Analyst

Professional Courses

- Chartered Financial Analyst (CFA)
- Certificate in Quantitative Finance (CQF)
- Financial Risk Manager (FRM)
- Certificate in Quantitative Risk Finance (CQRF)
- Indian Institute of Quantitative Finance also offer several certification programs.
- Masters in Financial Engineering offered by many leading universities.

Bonds Market

Definition (Bond)

A bond is a debt instrument issued by borrowers to obtain cash for either short term or long term needs.

Example (Issures of bonds)

- Sovereign governments
- State governments
- Corporations

Example (Buyers of bonds)

- Pension funds
- Insurance companies
- Foreign countries
- Individual investors

Purpose of issuing and buying bonds

Purpose of issuers

- Governments issue bonds to generate funds for constructing roads, expanding railways, spending on defense, providing education and for other expanses.
- Financial institutions issue bonds to invest in higher yielding assets and for expanding the business.

Purpose of buyers

- The main reason of investing in bonds is to have a diversify portfolio of investment.

Example (10 year US Treasury note)

1. Pricing date — July 22, 2005
2. Price — 99.214
3. Face value — 100
4. Market yield — 4.223%
5. Issuer — US Treasury
6. Maturity date — May 15, 2015
7. Coupon — 4.125%
8. Issued amount — $ 24.27 billion
9. Outstanding amount — $ 22 billion
10. Issue date — May 15, 2005

Example (General Motors bond)

1	Pricing date	Dec 29, 2005
2	Price	66
3	Market yield	13.299%
4	Issuer	General Motors
5	Maturity date	July 15, 2023
6	Coupon	8.25%
7	Issued amount	$ 1.25 billion
8	Outstanding amount	$ 1.25 billion
9	Issue date	June 26, 2003
10	Call	GM has the option to call
11	Rating	Noninvestment grade

The idea of interest rate is to compensate the lender for his money since money has time value. Thus, Rs 100 today is equivalent to Rs 100+**compensation** in future.

Further, the term **compensation** depends on many factors such as

- Duration of borrowing
- Trustworthiness of the party that is borrowing

India's ancient idea on interest

Chanakya's interest rate structure was risk-weighted. In his opinion if a person was doing business from safe way(less risk) charge him/her less and if he/she was doing business from forest or sea(risk increased) charge him/her more. In other words the rate of interest increased with the risk involved in the borrowers' business.

Cash-flows from a bond

1. Suppose, there is a bond with face value \$100 and coupon rate C and having a maturity of 1 year.
2. Suppose the coupon payment frequency is **semi-annual**.
3. The cash-flows from this bond include a coupon payment of $\$C/2$ after 6-months and a payment of $\$100 + C/2$ after 1-year.
4. If the coupon payment frequency is **quarter**.
5. In this case the cash-flows from this bond include a first coupon payment of $\$C/4$ after 3-months, a second coupon payment of $\$C/4$ after 6-months, a third coupon payment of $\$C/4$ after 9-months and a final payment of $\$100 + C/4$ after 1-year.

Discounted value of a cash-flow

- As we discussed earlier, Rs 100 received today is different than receiving Rs 100 after one year. In fact if prevailing one-year interest rate is 2% the value of Rs 100 to be received a year later will have value only Rs $100/1.02 = 98.04$ today.
- In other words if we have Rs 98.04 today we can invest this amount in bank and will receive Rs $98.04(1 + .02) = Rs100$ a year later. This Rs 98.04 today is called the **present value(PV)** of the Rs 100 a year later.
- The concept of **present value** is used extensively in finance for comparing two different cash-flows.

Convertible Bonds

A **Convertible Bond** is a hybrid fixed-income security that yields interest payments, but can also be converted into a predetermined number of **Common Stocks**.

- The price of convertible bond is sensitive to changes in interest rates.
- The **conversion ratio** determines how many shares you can get by converting one bond.
- If the stock price increase significantly, the investor can convert the bond to stock and hold or sell the stock whenever deem appropriate.

Source of Risks for Bond Holders

Interest rate risk (Market risk)

- This is the major factor that affects bond prices
- The price of bond changes in the opposite direction of interest rate change
- All bonds are exposed to interest rate risk

Why it happens: If you hold a bond paying 6% and the RBI suddenly raises interest rates so that new bonds pay 8%, your 6% bond becomes less attractive. To sell it, you must lower your price (sell at a discount) so the buyer's actual return (yield) matches the new 8% market rate.

Inflation risk

- Inflation reduces purchasing power that mean same amount of money has lesser value in period of high inflation.
- In other words same bond has different yields(real) in different inflation scenarios

Credit risk

- Inability of issuer to pay coupon and/or principal. For example
  - Corporate
  - Emerging market
  - High-yield bonds

Liquidity risk

- Inability to close position without substantial costs
- Some examples of illiquid bonds include municipal, corporate, and emerging market bonds

Term structure of interest rates (or yield curve)

Definition (Yield curve)

A yield curve is the relationship between spot rates and term to maturity. It is a measure of the market's expectations of future interest rates given the current market conditions.

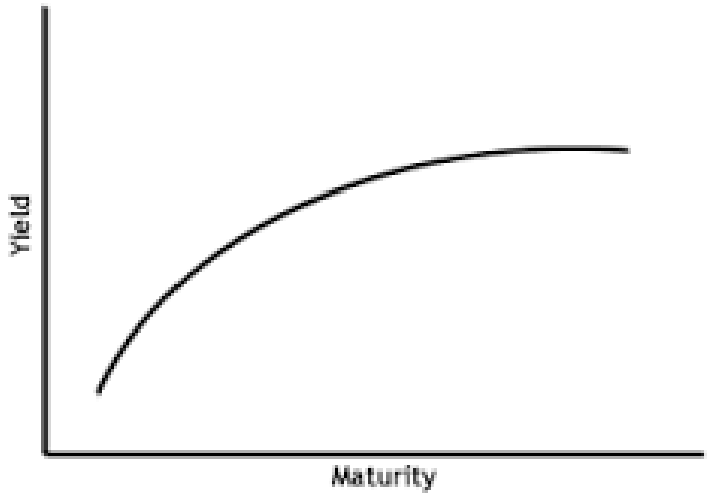


Figure: Normal Yield Curve

Credit spread Treasury and corporate bond

Since a Treasury bond is riskless and a corporate bond carry substantial credit risk, investors demand additional yield from corporate bonds. This yield difference is called **credit spread**.

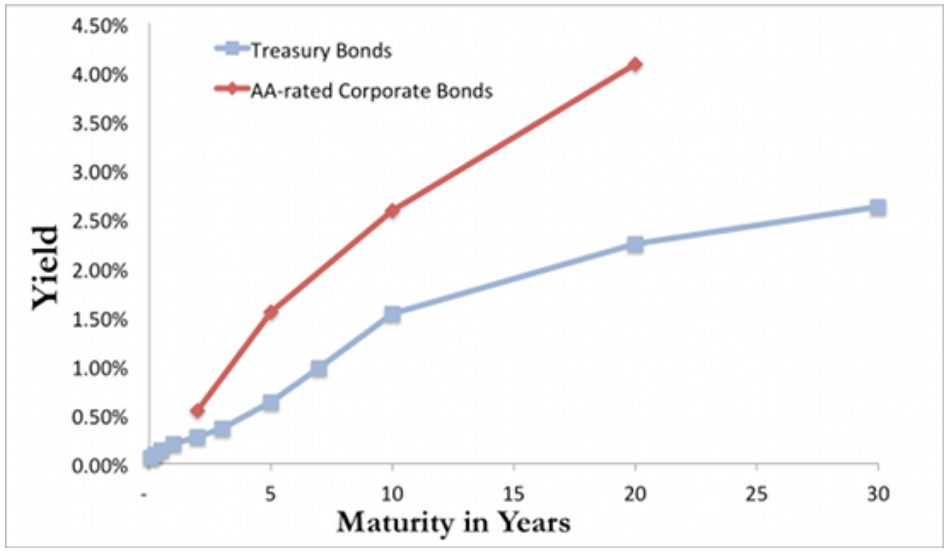


Figure: Credit Spread

There are many methods which have been used to model the yield curve. We can put them broadly into two categories; spline based models and function based models.

Spline Based Models

The idea of spline based models is to interpolate a spline function from the pre-known points. To get some features like continuous second derivative (which can be used in analysis of curvature) a cubic spline method is widely used.

Function based Models

The most popular function based models are **Nelson-Siegel**, and **Svensson models**. In fact, Svensson model is an extension of Nelson-Siegel model.

Stochastic Models

The most used interest rates models are: **Vasicek Model (1977)**, **Cox-Ingersoll-Ross (CIR) Model (1985)** and some others.

Vasicek Model

The Vasicek model describes the short rate dynamics by the following SDE:

$$dr_t = (\theta - ar_t)dt + \sigma dw_t,$$

where θ, $a > 0$ and σ is constant.

Cox-Ingersoll-Ross (CIR) Model

The CIR model describes the short rate dynamics by the following SDE:

$$dr_t = (\theta - ar_t)dt + \sigma\sqrt{r_t}dw_t,$$

where θ, $a > 0$ and σ is constant.

Stocks Market

Private Entity to Publicly Traded Firm

In August of 2004, Google went public, auctioning its shares in an unusual IPO format. The shares originally sold for $85 per share, and closed at over $100 on the first day. Currently Google shares are trading on Nasdaq at around $2442 per share.

Similarly, on May 18, 2012 Facebook went public with its IPO share value $42.05. Currently after 9 years its paer share value is roughly $350.

Stock

- Stock is an ownership interest in the company.
- When you buy stock, you receive “shares” of that company.
- A stock holder can receive the ownership certificate from the company.
- Provide right to vote for directors and on certain issues.

Figure: Sample Stock Certificate

Two Types

Common stock

- Right to vote.
- Receive dividends.

Preferred stock

- Receive a fixed dividend.
- Do not usually vote.
- Preferred shares are a hybrid securities issued by corporations marrying certain traits of equity and fixed income investment.
- Because preferred shares pay steady dividends, they will typically trade in the market for a value different from the same firm's common stocks.

Some US Equities from 2008 to 2018

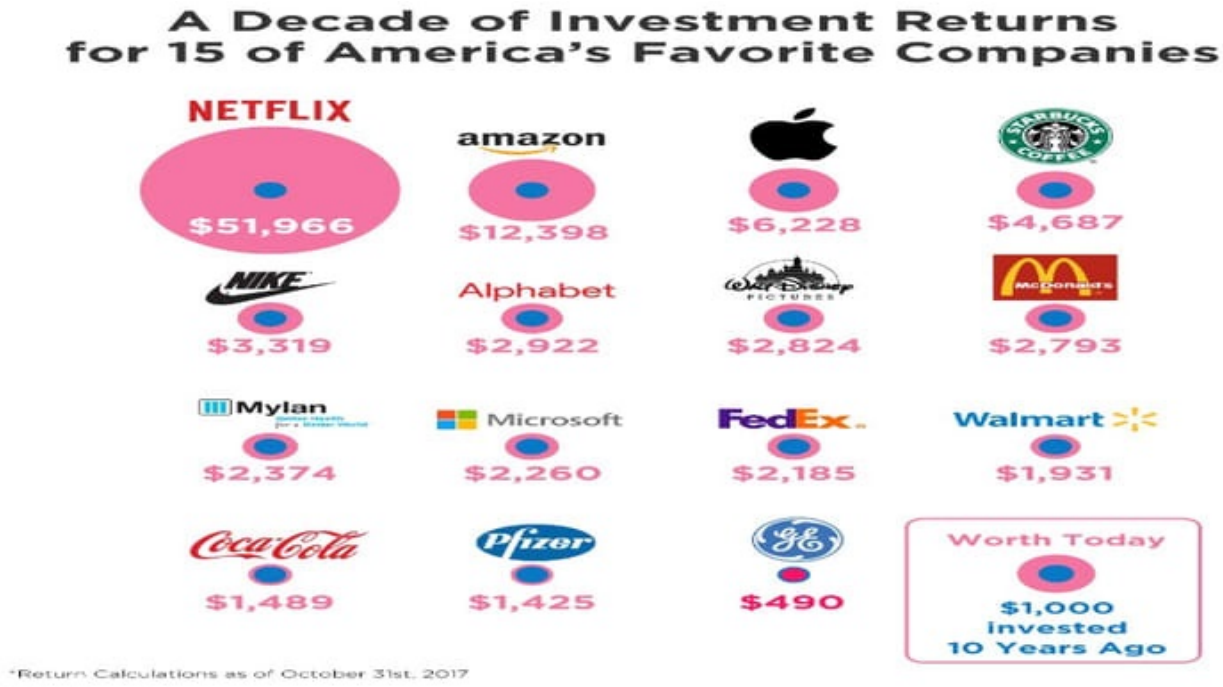


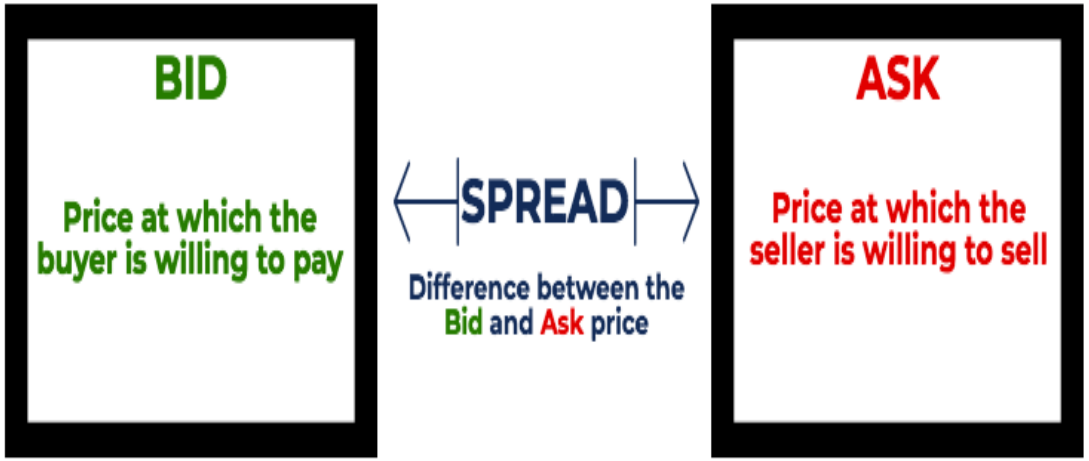
Figure: Credit Spread

Netflix

Figure: Netflix

How Share Prices Are Set ?

- The most common way to set share price is through an auction process where buyers and sellers place bids and offers to buy or sell.
- A bid is the price at which somebody wishes to buy, and an offer (or ask) is the price at which somebody wishes to sell.
- When the bid and ask coincide, a trade is made.
- Matching buyers and sellers of stocks on an exchange was initially done manually, but it is now increasingly carried out through computerized trading systems.



Face Value vs Market Value

Face Value: is the nominal value of the shares, that is their original cost as mentioned in share certificate. For example the TCS shares face value is Rs. 1, however it is currently trading close to Rs 3890.

Market Value: is the traded value of a stock governed by market forces like demand and supply.

- At the time of **stock split** both the face value and market values are adjusted using the split proportion.
- When a company announces a **dividend**, it is issued against the face value rather than the market price.
- If a company with face value Rs 1 and Market value Rs 1000 issue a dividend of 1000%, it means a dividend of Rs 10 per share.

Role of Stock Exchange

- Stock exchanges are the markets, where existing owners of shares can transact with potential buyers.
- If a stock is listed on exchange it provide liquidity for shares held by the company's shareholders.
- It enables the company to raise additional funds by issuing more shares.
- It provide greater visibility.
- Derivatives (like stock options) can be easily setup if a stock is listed on an exchange.

How do you make money on stocks?

An investor can make money on stock in two ways

- Dividends: which will be part of the company's profits (if any).
- Capital gains: which is selling your stock for more than what you paid for it.

Stock Index

- Stock market indexes are frequently used to monitor the behavior of a groups of stocks.
- They help us interpret investment results and provide a benchmark for performance measurement.
- Major indexes include the Dow Jones Industrial Average, the S&P 500, NASDAQ composite, Wilshire 5000, FTSE 100, FTSE 250, Nikkei 225 and others.
- Major index India include Nifty 50, BSE Sensex and Nifty Bank.

Index Construction

- Price weighted index
- Equal weighted index
- Capitalization weighted index

Price Weighted Index Construction

	Day One		Day Two
Company A	$30	→ 3-for-1 split →	$10
Company B	20	→	20
Company C	10	→	10
Total	$60	→	$40

Apparent change in index = –33.3%

Divisor		3.0			2.0
$60/3.0	=	$20	$40/2.0	=	$20

Apparent change = 0.0%

The divisor maintains economic integrity.

Equal Weighted Index Construction

Security	Initial Price	Subsequent Price	Return	Weighted Return
A	$ 10	$ 10.50	5.00 %	1.00 %
B	20	21.375	6.875	1.375
C	35	33	–5.71	–1.143
D	27	28.50	5.56	1.111
E	109	114	4.587	0.917
Index	100.00	103.26		3.260%

Market Capitalization Weighted Index Construction

Base Value

Company	Shares Outstanding	Price	Shares × Price
A	200,000,000	$30	$6,000,000,000
B	50,000,000	20	1,000,000,000
C	100,000,000	10	1,000,000,000
		Base value →	$8,000,000,000

Initial Value of the Index

$$\frac{(200{,}000{,}000 \times \$30) + (50{,}000{,}000 \times \$20) + (100{,}000{,}000 \times \$10)}{\$80{,}000{,}000^*} = 100.00$$

*Note: The arbitrary division by $80,000,000 keeps the index a workable size.

Later

Company	Shares Outstanding	Price	Shares × Price
A	200,000,000	$31	$6,200,000,000
B	50,000,000	19.50	975,000,000
C	100,000,000	10.25	1,025,000,000
			$8,200,000,000

New Value of the Index

$$\frac{\$8{,}200{,}000{,}000}{\$80{,}000{,}000} = 102.50$$

FX Market

Currency Market

- Currency is a medium of exchange for goods and services.
- In short, it is money, in the form of paper or coins, usually issued by a government and generally accepted at its face value as a method of payment.
- According to *WorldAtlas.com*, 180 national currencies recognized by the United Nations are currently in circulation.
- Another 66 countries either use the U.S. dollar or peg their currencies directly to the dollar.

Fixed exchange rates

A fixed exchange rate, also known as the pegged exchange rate, is “pegged” or linked to another currency or asset (often gold) to derive its value. Such an exchange rate mechanism ensures the stability of the exchange rates by linking it to a stable currency itself. Also, a fixed currency system is relatively well protected against the rapid fluctuations in inflation. Some countries following a fixed rate system include Denmark, Hong Kong, Bahamas & Saudi Arabia.

Advantage: A country with a fixed exchange rate system is attractive to foreign investors who are lured to invest in that country due to the stability it offers.

Disadvantage: The government of a country following such a system has to maintain a huge amount of foreign or gold reserves to maintain its value. This system thus proves to be an expensive one.

Floating Exchange Rate

Floating exchange rate systems are ones whereby the rate of a currency is determined by the market forces of demand and supply. Unlike the fixed exchange rate they do not derive their value from any underlying. Some economists argue that a floating system is more preferable since it absorbs the shocks of a global crisis and automatically adjusts to arrive at an equilibrium.

Advantage: The rates under this system are determined by a self sufficient mechanism. Therefore, the dependence on government or international monetary organizations is minimum.

Disadvantage: Floating rate systems are prone to greater volatility since they are determined by the market forces. The increased volatility increases the risk quotient in such markets consequently making it a relatively expensive place for the foreign investors.

Factors for FX Market

Factors that affect the FX Return are termed as Signals . They can be interest rates of bank, monetary policy, political stability, imports and exports of that country.

Short Term Rate: Short term rate is interest rate for a period like 1 month and 3 months. Short term interest rates worldwide are administered by the nations' central banks. In the United States, the Federal Reserve Board's Open Market Committee (FOMC) sets the federal funds rate, the benchmark for other short term interest rates.

Long Term Rate: Long term stands for denoting that it is for long period say, one year or 2 or 3 years, or 10 years and upto 30 years. While short term interest rates are administered by central banks, long term interest rates are determined by market forces.

Term Spread: The difference between short term and long term rates.

PPP Based Exchange Rates: Purchasing power parity based exchange rates are obtained by comparing two currencies based on a basket of goods and services.

Derivatives

Definition (Derivatives)

Derivatives are financial securities whose payoffs are explicitly given by the value of some other underlying asset.

Examples

Important examples of derivatives are forward contracts, futures contracts, options, swaps, swaptions.

- The derivatives market is often estimated at more than $1.2 quadrillion (1200 trillion).
- Some reports says that the derivatives market is more than 10 times the size of the total world GDP.

Forward Contracts

Forward contract on an asset is a contract to buy or sell a unit amount of the asset at a specific price and at a fixed time in the future.

Example

Consider a Forward contract to buy 100,000 kg of sugar at 40 Rs per kg on November 15, 2024. Some features of Forward contracts:

- The contract is between two parties.
- The asset involved in the forward contract is called the underlying asset.
- Buyer is said to be in long position of the Forward contract.
- Seller is said to be in short position.
- The agreed price of the asset in the contract is called the delivery price.
- The time of trading of the asset in the contract is called maturity.
- The market for immediate delivery of the asset is called the Spot market.
- The price of the asset in the spot market is called Spot price.

Definition (Swap)

Swap is a contract to exchange one cash flow for the other.

Example

The most common kind of swap is an interest rate swap. Swap trades on OTC markets. In an interest rate swap, the parties exchange cash flows based on a notional principal amount (this amount is not actually exchanged) in order to hedge against interest rate risk or to speculate.

Futures Contract

Futures contract is like forward contract with the difference of being traded in an organized exchange.
Futures contract is designed to eliminate the default risk associated with forward contract.
It is achieved through the process called marking to the market.

- While entering into the futures contract both parties need to deposit an amount with the broker called the margin account.
- At the end of each day, the net receipt will be marked. i.e. at the end of each day, the delivery price will be revised to the current delivery (futures price) and the net receipt from this process is added to the market.
- For example, at the end of day one, for the party holding the long position in the futures contract $F_1 - F_0$ will be deposited in her/his margin account, where $F0$ is the futures price at the time of entry of the contract and F_1 is the futures price at the end of day one.
- when the margin value falls below a certain level, the contract holder need to pay money to makeup the margin account.

Definition (Option)

An option is a financial contract where the buyer of the contract has the right to buy/sell specific number of units of the underlying asset at a specified price on or before a specified date. This right is granted by the seller of the option.

Terminologies related to options

- Buyer of the option is called holder and seller is called writer.
- The specific time till the option is valid is called the maturity.
- Specified price = **strike** or **exercise price**
- Buyer = holder = long position
- Seller = writer = short position
- If the holder has the write to buy, it is called **call option**.
- If the holder has the write to sell, it is called **put option**.
- On/before: **American**. Only on expiration: **European**.

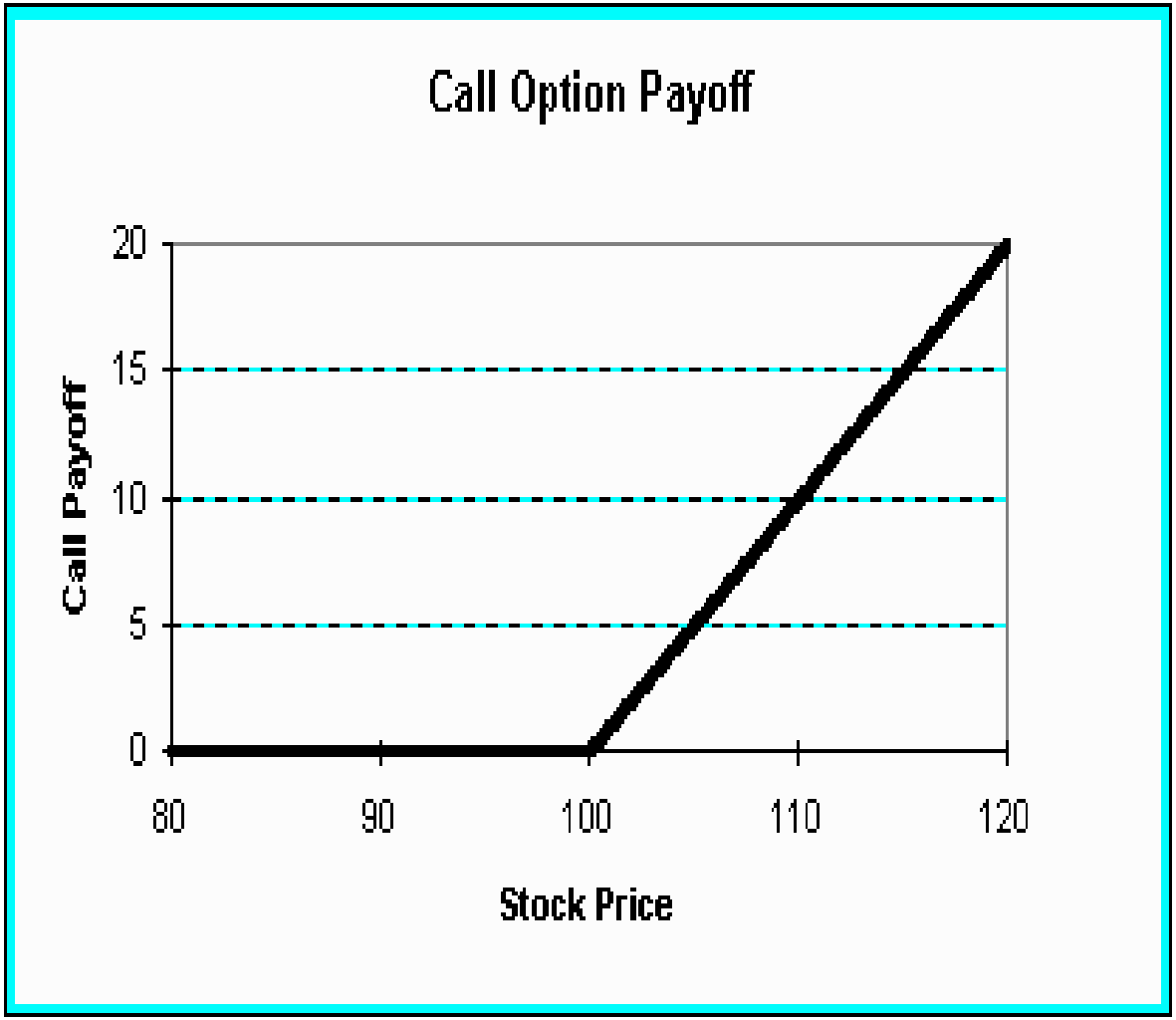
In the Money and Out of the Money Options

- A call option with strike K is said the be **in the money option** if $S_0 > K$.
- Call option is called **out of the money** if $S_0 < K$.
- It is called **at the money** if $S_0 = K$.
- Here S_0 is the current price of the underlying asset.
- The payoff at maturity of a European call option is

$$(S_T - K)^+ = \max\{S_T - K, 0\},$$

where S_T is the price of the underlying at the maturity T.

Call Option Payoff



Factors which Affect Option Price

- **Current price of the underlying:** bigger the current price bigger the call option price.
- **Strike price:** bigger the strike price smaller the call option price.
- **Volatility:** high volatility implies high option price for both call and put.
- **Maturity:** large time to maturity imply high option price.

Covered Call

Covered Call: Portfolio of a long position in the underlying asset and a short position in the call option.

- The main goal of the covered call is to collect income via option premiums by selling calls against a stock that you already own.
- Assuming the stock doesn't move above the strike price, you collect the premium and maintain your stock position.
- Covered call writing is typically used by investors and longer-term traders, and is rarely used by day traders.

Example

Suppose you own 100 shares of XYZ corp. XYZ corp 3-months call option price with strike 110 is Rs 5. A covered call strategy in this scenario would be to have a short position in the call option. Covered call are generally employed in OTM options.

Bull Spread

Portfolio of a long position in call option with strike K_1 and a short position in call option with strike K_2, where $K_1 < K_2$. Payoff at maturity T is given by

$$\begin{cases} 0 & \text{if } S_T \leq K_1 \\ S_T - K_1 & \text{if } K_1 < S_T \leq K_2 \\ K_2 - K_1 & \text{if } S_T > K_2. \end{cases}$$

If **both the calls are OTM**, then

- Cost of buying the bull spread is low.
- Small chance of receiving the maximum payoff $K_2 - K_1$.

If **both call are ITM**, then

- Cost of buying the bull spread is high.
- High probability of receiving the maximum payoff $K_2 - K_1$.

Bear Spread

Portfolio of a short position in put option with strike K_1 and a long position in put option with strike K_2, where $K_1 < K_2$. Payoff at maturity T is given by

$$\begin{cases} K_2 - K_1 & \text{if } S_T \leq K_1 \\ K_2 - S_T & \text{if } K_1 < S_T \leq K_2 \\ 0 & \text{if } S_T > K_2. \end{cases}$$

If **both the puts are OTM**, then

- Cost of buying the bear spread is low.
- Small chance of receiving the maximum payoff $K_2 - K_1$.

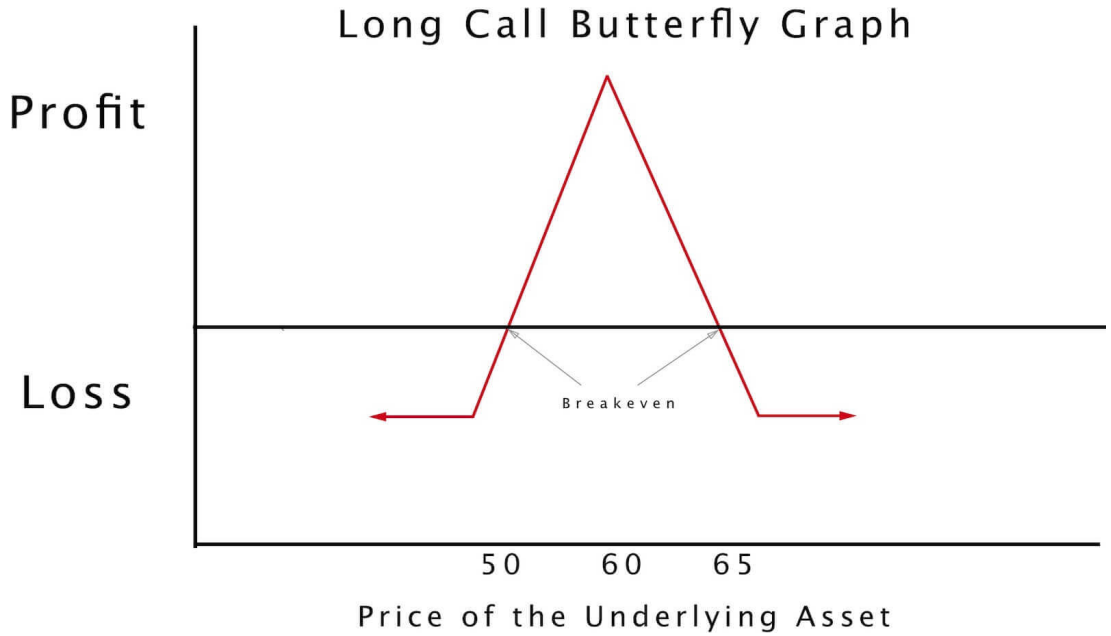
If **both puts are ITM**, then

- Cost of buying the bear spread is high.
- High probability of receiving the maximum payoff $K_2 - K_1$.

Butterfly Spread

Portfolio of long positions in two call options with strikes K_1, K_3 and two short positions in a call option with strike K_2, where $2K_2 = K_1 + K_3$. Payoff at maturity T is given by

$$\begin{cases} 0 & \text{if } S_T \leq K_1 \\ S_T - K_1 & \text{if } K_1 < S_T \leq K_2 \\ 2K_2 - K_1 - S_T & \text{if } K_2 < S_T \leq K_3 \\ 2K_2 - K_1 - K_3 = 0 & \text{if } S_T > K_3. \end{cases}$$



Definition (Arbitrage)

The simultaneous buying and selling of a security at two different prices in two different markets, resulting in profits without risk. Perfectly efficient markets present no arbitrage opportunities. Perfectly efficient markets seldom exist, but, arbitrage opportunities are often precluded because of transactions costs.

Example

As a hypothetical example, if Infosys stock trades at Rs 790 on the National Stock Exchange (NSE) and at Rs 800 on the Bombay Stock Exchange (BSE), an investor could guarantee a profit by purchasing the stock on the NSE and simultaneously selling the same amount of stock on the BSE.

Caps

The Cap guarantees a maximum rate to the buyer. Borrowers are interested by caps since they set a maximum paid interest cost. A cap is an option which has value only when the rate is above the guaranteed rate, otherwise, it is worthless.

Floors

The floor guarantees a minimum rate to the buyer. Investors are interested by floors because they are willing to be protected against declining interest revenues. A floor is an option and it has value only when the rate is below the guaranteed rate, otherwise, it is worthless.

Collars

A collar combines a cap and a floor. The buyer of a collar buys a cap from the bank and sells a put to the bank. The premium received from the sale of the put reduces the cost of the straight cap. The buyer is typically a borrower floating rate.

Factors for Cross Sectional Study

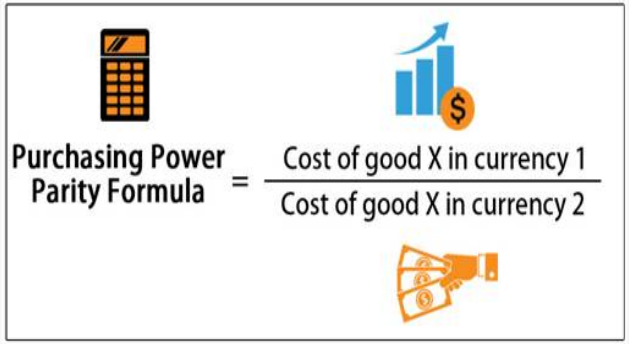
Carry Factors

- Carry factors are based on the opportunities to borrow and pay interest in order to buy something else that has higher interest.
- For example, in currency market it is to borrow money in one currency at a low interest rate and lends the same money in other currency at a higher interest rate.
- In equity market carry is the difference between the dividend yield and funding costs to buy the equity

.

Value Factor

- Valuation factors are based on the deviation of the price of an asset from its fundamental price.
- For example, in case of currency market theoretical price is the purchasing power parity (PPP) rate and market price is the current FX rate.



Momentum Factors

- Momentum factors is the analysis of financial market movements.
- The idea is to look at price changes that occur on period-to-period basis.
- As discussed in the literature buying winners and selling losers generate significant positive returns over 3- to 12-month holding period (Jegadeesh and Titman, 1993, The Journal of Finance).

Economic Factors

- Economic factors are based on overall analysis of the market by analyzing the interplay of macroeconomic and micro-economic variables.
- Such as interest rates, inflation, monetary policies etc.
- These variables affects the demand and supply of an asset and hence their prices.

Cross Market Factors

- Cross market factors are based on the inter-relationships of different markets.
- For example equity and commodity market are poorly correlated.
- Exchange rates are directly related to the interest rates in respective countries.
- Also, it is known that the difference between bonds and equity market returns can be predicted by using the term structure of forward rates

Feature Engineeing

122.639999, 122.510002, 118.019997,116.629997, 116.239998, 118.160004,

```
def features(dataset, timestep):
    datax = []
    datay = []
    a = []
    print(len(dataset))
    for i in range(0 , len(dataset) - timestep ):
        #print(i)

        a = list(dataset['Close'][i:i+timestep])
        #print(a)
        datax.append(a)
        b = dataset['Close'][i+timestep]
        #print(b)
        datay.append(b)
    return ( datax, datay)

X_train1, y_train1 = features(X_train, 3)
X_test1, y_test1 = features(X_test,3)
```

	f1	f2	f3	close
0	122.639999	122.510002	118.019997	116.629997
1	122.510002	118.019997	116.629997	116.239998
2	118.019997	116.629997	116.239998	118.160004
3	116.629997	116.239998	118.160004	117.330002
4	116.239998	118.160004	117.330002	117.110001
5	118.160004	117.330002	117.110001	119.120003
6	117.330002	117.110001	119.120003	116.709999
7	117.110001	119.120003	116.709999	114.379997
8	119.120003	116.709999	114.379997	109.959999
9	116.709999	114.379997	109.959999	107.660004

Cross Sectional Study Contradicting EMH

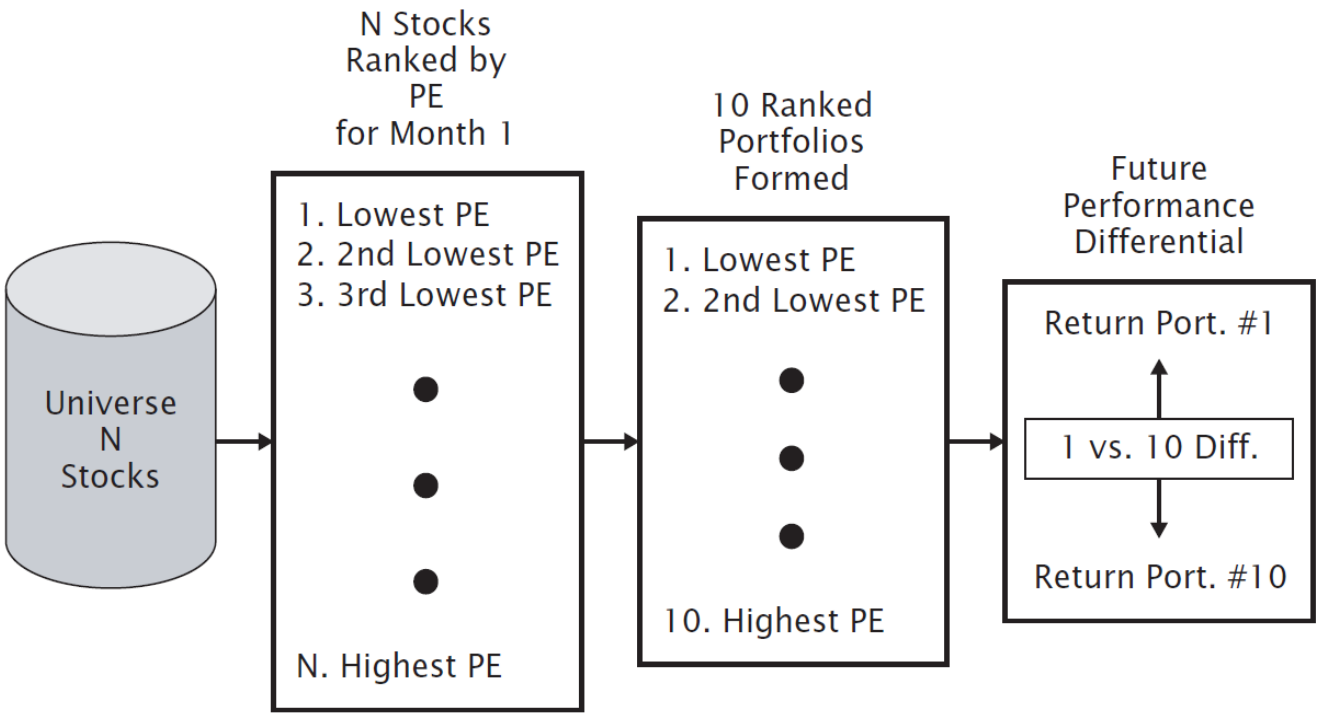


Figure: Cross Sectional Time-Series Analysis

[1]

[1] Aronson, D.R. (2007). Evidence-Based Technical Analysis, John Wiley & Sons, New Jersey.

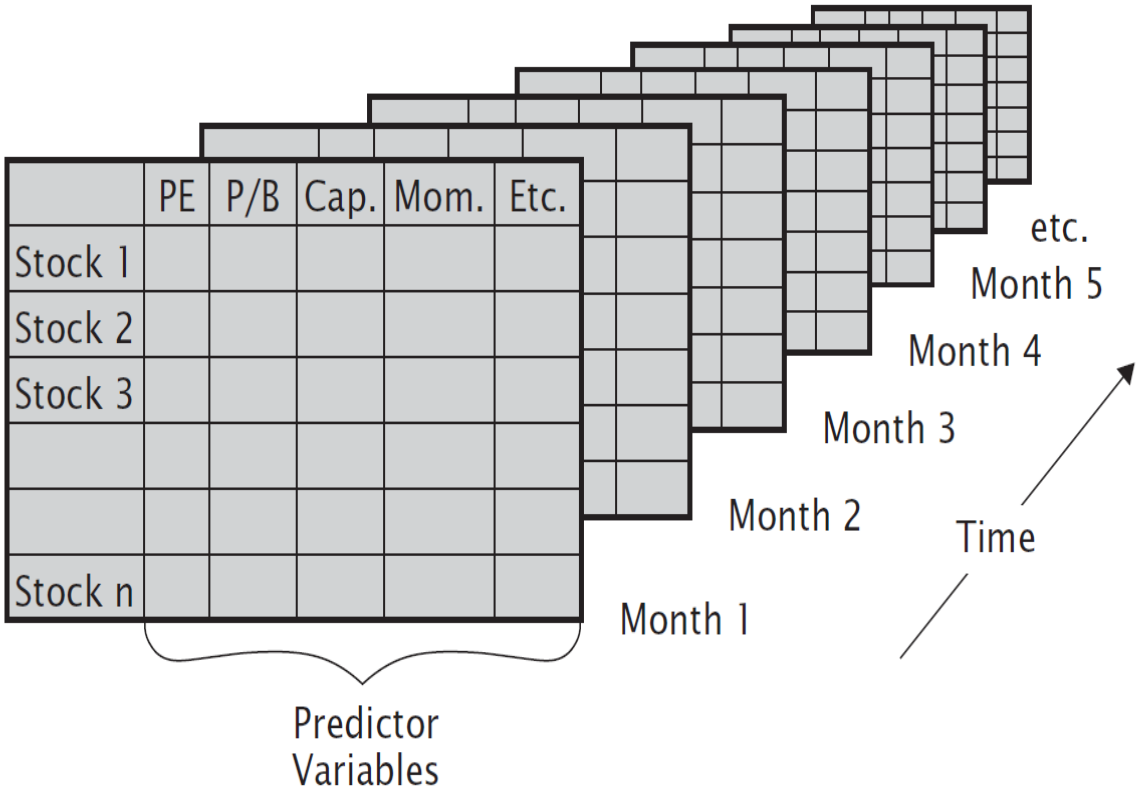


Figure: Cross Sectional Time-Series Analysis

2

[2] Aronson, D.R. (2007). Evidence-Based Technical Analysis, John Wiley & Sons, New Jersey.

Features Engineering Continued

Time-series z-score: Calculates the time-series z-score of a factor based on a look-back period (e.g. 5-year). The input to this function is a time series and a look-back parameter.

Change: Calculates the change in the factor value based on a look-back period. The input to this function is a time-series of factor values and a look-back period.

Moving-average: Calculates the moving average of a factor time-series based on given time intervals. For example the carry1-12 is calculated as the moving average of last 12-months carry including the most recent month. Again, here the inputs are a time-series of factor values and a look-back period.

Deviation from historical average: Calculates the current factor deviation from its long-term average. Mathematically, it is written as $d_t = C_t - \frac{(C_{t-\Delta}+C_{t-2\Delta}+\cdots+C_{t-N\Delta})}{N}$, where N denotes the number of look-back periods and Δ denotes the period length at which data is available.

Factors Dataframe

x - DataFrame

index	10Yr	SMA	EMA	Returns	Momentum1	Momdiv5	Momentum5	Ret	RSI	Term Spread
2013-10-16	.35295	-1.73142	-1.73421	-0.0412099	0.150518	0.636387	0.318808	0.148971	0.786873	1.44978
2013-10-17	.11597	-1.72062	-1.72358	-0.040295	0.202222	0.915861	0.481667	0.208085	0.290048	1.34699
2013-10-18	.08635	-1.70465	-1.71144	-0.0390352	0.256751	0.990314	0.529651	0.269363	0.503943	1.34699
2013-10-21	.17522	-1.68806	-1.69252	-0.0170373	0.695697	1.3386	0.747012	0.767423	1.20107	1.41552
2013-10-22	.908613	-1.66976	-1.67722	-0.0418639	-0.062845	1.08439	0.598959	-0.095148	1.0285	1.27846
2013-10-23	.819745	-1.65128	-1.66058	-0.0381871	0.294905	1.23579	0.694158	0.303837	1.32625	1.20993
2013-10-24	.87899	-1.62876	-1.64153	-0.0348632	0.395788	1.43607	0.821781	0.412336	1.79473	1.22706
2013-10-25	.87899	-1.61072	-1.62909	-0.0369611	-0.304938	0.818713	0.451444	-0.357414	1.26113	1.24419
2013-10-28	.908613	-1.59391	-1.6157	-0.0398828	0.231116	0.307716	0.145134	0.2299	1.65353	1.26133
2013-10-29	.87899	-1.58497	-1.61299	-0.01621	-0.69848	-0.359353	-0.273957	-0.789659	0.76808	1.24419
2013-10-30	.938235	-1.57316	-1.60513	-0.0316067	0.464655	-0.1808	-0.162036	0.496352	0.98778	1.27846
2013-10-31	.99748	-1.56411	-1.59981	-0.0414761	-0.100836	-0.691201	-0.489473	-0.136531	0.824347	1.32985
2013-11-01	1.01685	-1.55858	-1.59699	-0.0411132	-0.126797	-0.511845	-0.37202	-0.165899	0.557785	0.0963301
2013-11-04	1.2242	-1.5559	-1.59005	-0.0352105	0.383297	-0.352353	-0.271683	0.402354	0.651233	0.0106687
2013-11-06	3.0985511	-1.55387	-1.58591	-0.0370749	-0.298312	0.241884	0.101375	-0.35357	0.345499	0.627431
2013-11-07	.08635	-1.55857	-1.58799	-0.0312389	-0.439414	-0.70912	-0.496469	-0.514622	-0.0508043	1.32985
2013-11-12	.85653	-1.56489	-1.58248	-0.0420643	0.0705712	-0.00447518	-0.0514335	0.0535755	-0.340032	1.77529
2013-11-13	.53068	-1.56797	-1.57954	-0.0421479	0.052214	-0.350105	-0.269594	0.0329865	-0.267575	1.58684
2013-11-14	.35295	-1.56073	-1.57195	-0.0334516	0.427072	0.235508	0.101031	0.450663	-0.187111	1.48404

Format Resize Background color Column min/max Save and Close Close

Time-Series and ML Techniques

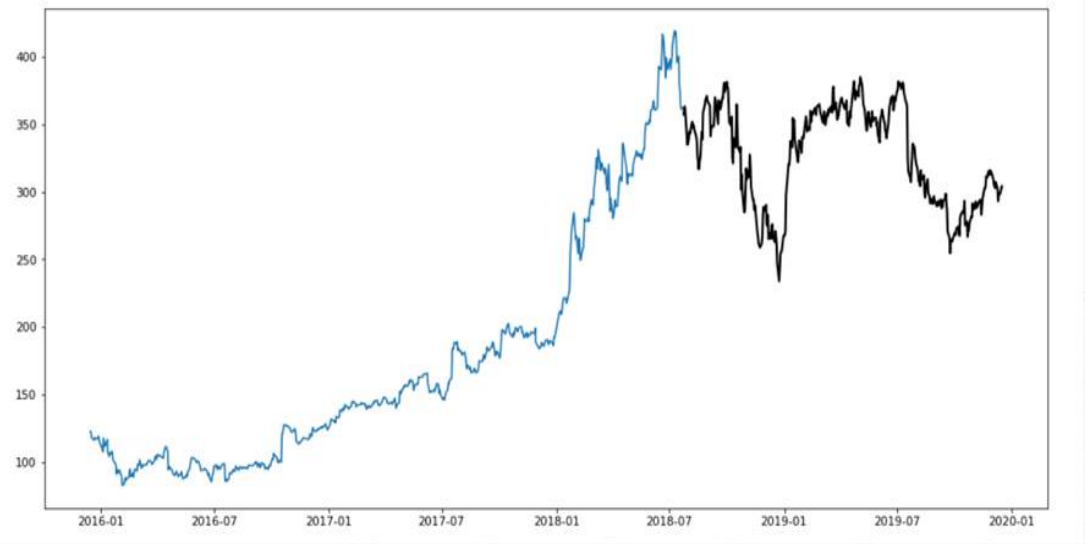
- Once features and response variables data are available, we use the statistical techniques to train/estimate the model.
- We use 70% data to for training part and remaining for testing part.
- We have used Panel Regression, Time-Series Regression techniques for our analysis.
- Further, we have used Logistic Regression, LDA, QDA, Decision Trees, SVM Techniques for our prediction purpose.
- We have to explore Deep Learning Techniques like LSTM.

Train and Test Split

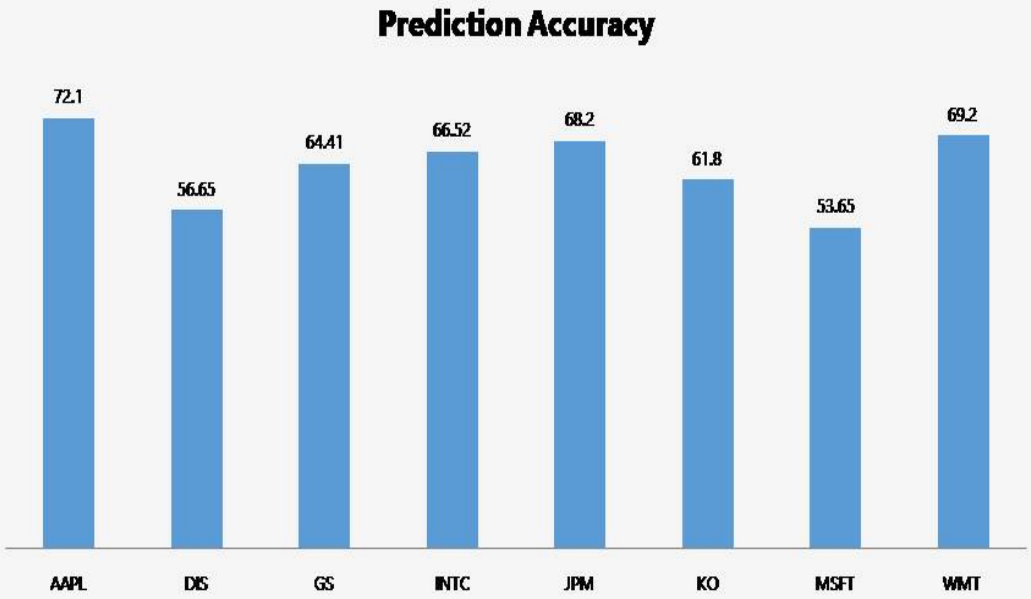
Divide the Data

Generally (65-70)% of the data is used to train the model.

(30 -35) % of the data is used to predict the prices and calculate the accuracy of the model.



Test Prediction Accuracy For Some Equities



Technical Analysis

What is Technical Analysis ?

Technical analysis is the study of price movements and changes in trading volume to predict future movements of stock prices.

- Generally investors look at the history of prices and volume, and they analyze how such prices and volume behave in order to predict if prices will increase or decrease.
- Technical analysis is generally focused on the prediction of direction of the market rather than the absolute prices.

Equity Market and Fashion

- According to *Skirt Length Theory* if skirts are short, it means the markets are going up.
- And if the skirt is long, it means the markets are heading down. Also called the Hemline Theory.

3

[3] https://socionomics.net/

Data Format for Technical Analysis

Index	High	Low	Open	Close	Volume	Adj Close
2014-12-31 00:00:00	531.142	524.36	529.795	524.959	1368246	524.959
2015-01-02 00:00:00	529.815	522.665	527.562	523.373	1447563	523.373
2015-01-05 00:00:00	522.894	511.655	521.827	512.463	2059840	512.463
2015-01-06 00:00:00	514.762	499.678	513.59	500.586	2899940	500.586
2015-01-07 00:00:00	505.855	498.282	505.612	499.728	2065054	499.728
2015-01-08 00:00:00	502.101	489.656	496.627	501.304	3353582	501.304
2015-01-09 00:00:00	503.538	493.435	503.378	494.811	2071371	494.811
2015-01-12 00:00:00	494.618	486.225	493.585	491.201	2326771	491.201
2015-01-13 00:00:00	501.603	491.042	497.474	494.821	2370490	494.821
2015-01-14 00:00:00	501.852	491.65	493.296	499.499	2235721	499.499
2015-01-15 00:00:00	504.295	496.397	504.186	500.416	2715836	500.416
2015-01-16 00:00:00	506.799	498.631	498.641	506.689	2298293	506.689
2015-01-20 00:00:00	511.097	504.631	509.601	505.512	2232011	505.512
2015-01-21 00:00:00	517.858	504.814	505.861	516.622	2268712	516.622
2015-01-22 00:00:00	534.862	518.277	520.052	532.927	2676929	532.927
2015-01-23 00:00:00	540.686	531.541	534.124	538.472	2281747	538.472
2015-01-26 00:00:00	537.524	528.22	537.056	533.745	1543727	533.745
2015-01-27 00:00:00	529.247	516.771	528.519	517.21	1904013	517.21
2015-01-28 00:00:00	521.558	508.604	521.349	508.604	1683810	508.604
2015-01-29 00:00:00	509.691	499.828	509.601	509.262	4186362	509.262
2015-01-30 00:00:00	538.392	514.109	514.448	533.057	5606350	533.057
2015-02-02 00:00:00	531.541	517.13	530.274	527.033	2849803	527.033
2015-02-03 00:00:00	531.94	521.827	526.554	527.791	2038682	527.791
2015-02-04 00:00:00	531.214	519.843	527.791	521.329	1663655	521.329
2015-02-05 00:00:00	527.053	520.661	522.356	526.135	1849765	526.135
2015-02-06 00:00:00	535.729	524.969	526.195	529.546	1763529	529.546

Price Fields for Technical Analysis

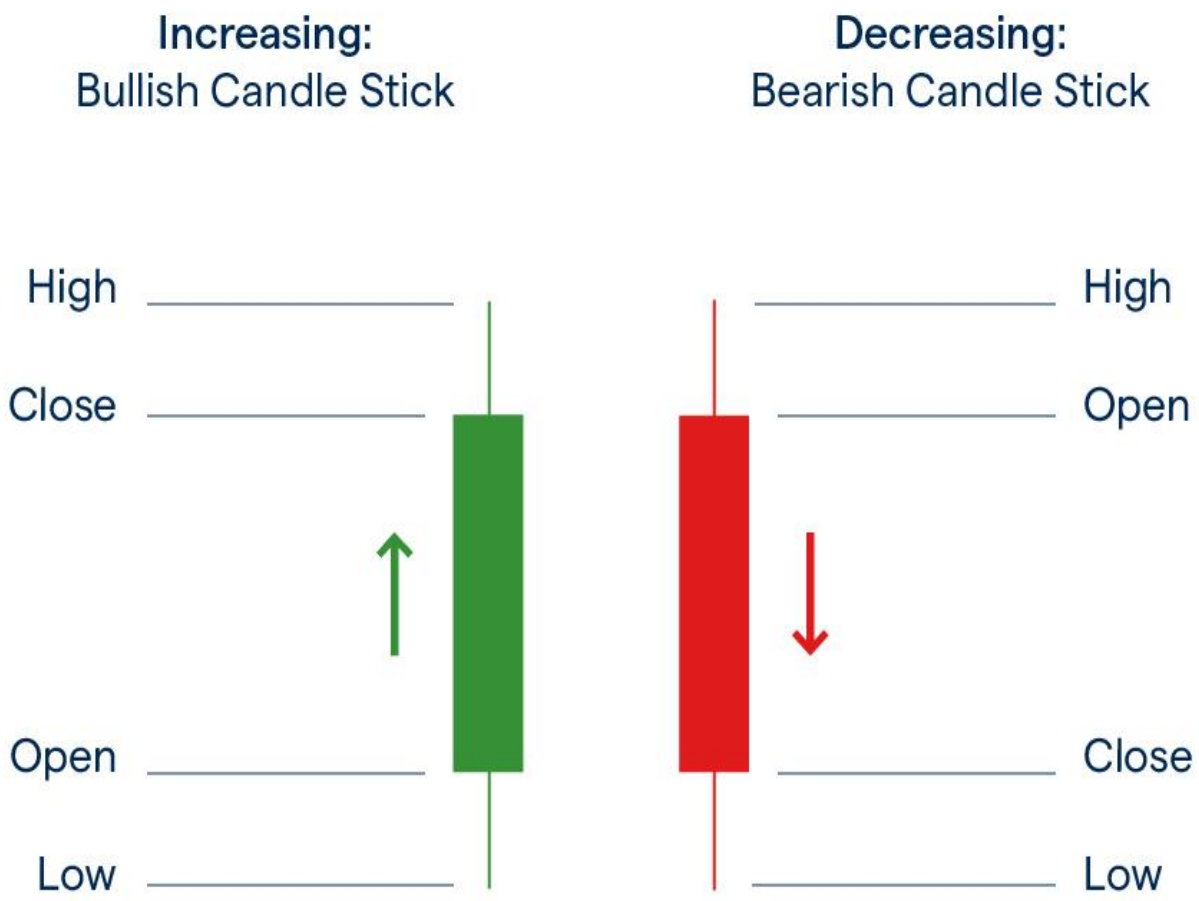
- Open: This is the price of the first trade for the period (e.g., the first trade of the day).
- High: This is the highest price that the security traded during the period.
- Low: This is the lowest price that the security traded during the period.
- Close: This is the last price that the security traded during the period. The Close is the most often used price for analysis.
- Volume: This is the number of shares (or contracts) that were traded during the period. Generally increasing prices accompanied with increasing volume.
- Open Interest: This is the total number of outstanding contracts.
- Bid: This is the price a market maker is willing to pay for a security (i.e., the price you will receive if you sell).
- Ask: This is the price a market maker is willing to accept (i.e., the price you will pay to buy the security).

Line Chart

Google Equity Prices
2500
2000
1500
1000
500
Close
2015
2016
2017
2018
2019
2020
2021
Date

Candle Stick Chart

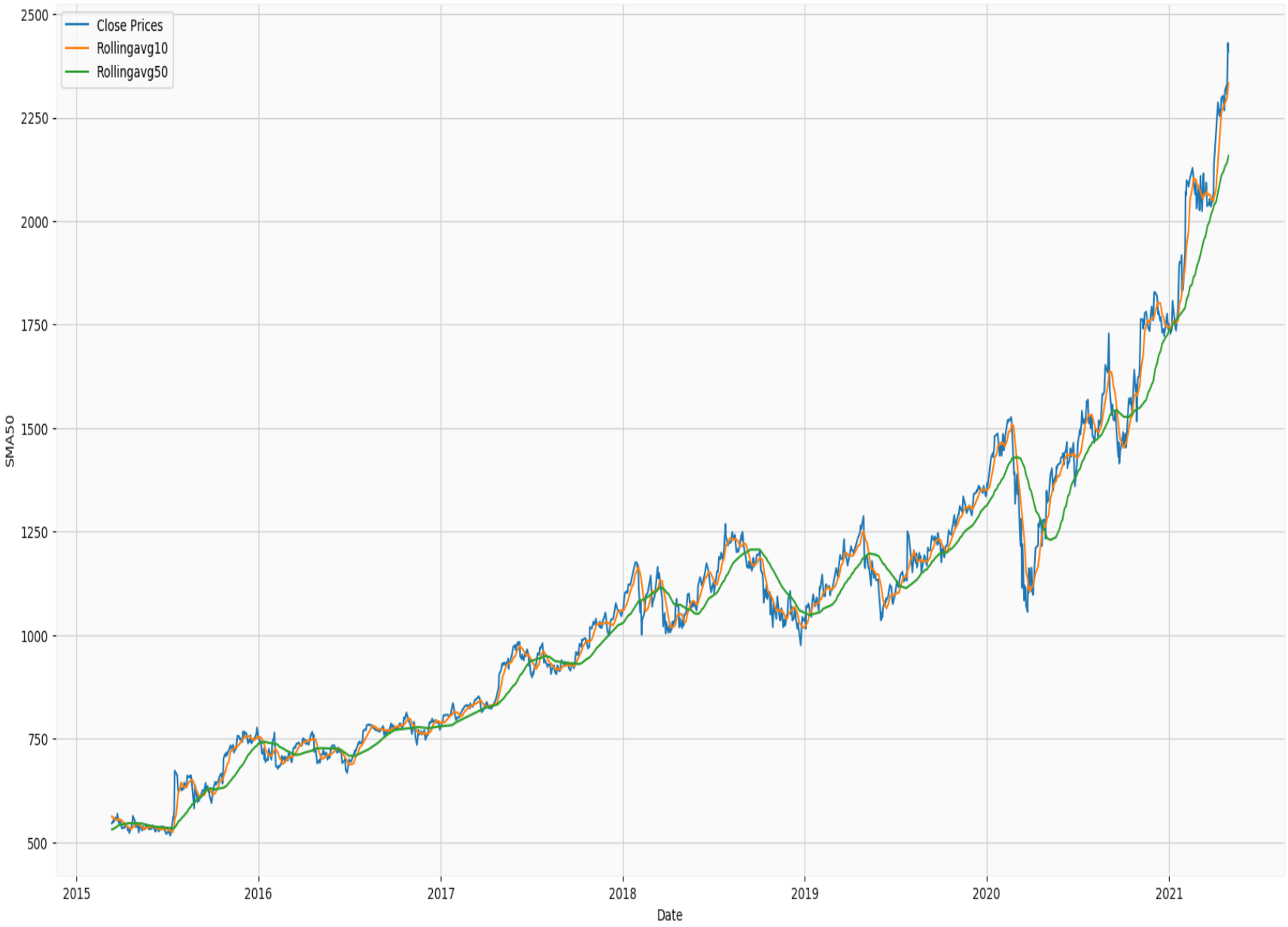
The candlestick chart is a style of financial chart describing open, high, low and close for a given x coordinate (most likely time). The boxes represent the spread between the open and close values and the lines represent the spread between the low and high values.



IG.COM

Candlestick Chart

Rolling Average Chart for Google Equity



A simple Trading Strategy Based on Simple Moving Average

- "Buy" arrows were drawn on the chart when equity price rose above its 200-day moving average.
- Further, the "sell" arrows were drawn when price fell below its 200-day moving average.

Channel Breakout Operator (CBO)

- The lower boundary is defined by the minimum value of the time series over the past n-periods, not including the current period.
- The upper boundary is defined by the time series' maximum value over the past n-periods, not including the current period.
- The channel breakout operator signals long positions if the analyzed series exceeds its maximum value established over the past n-periods.
- Conversely, short positions are signaled when the time series falls below its n-period minimum.

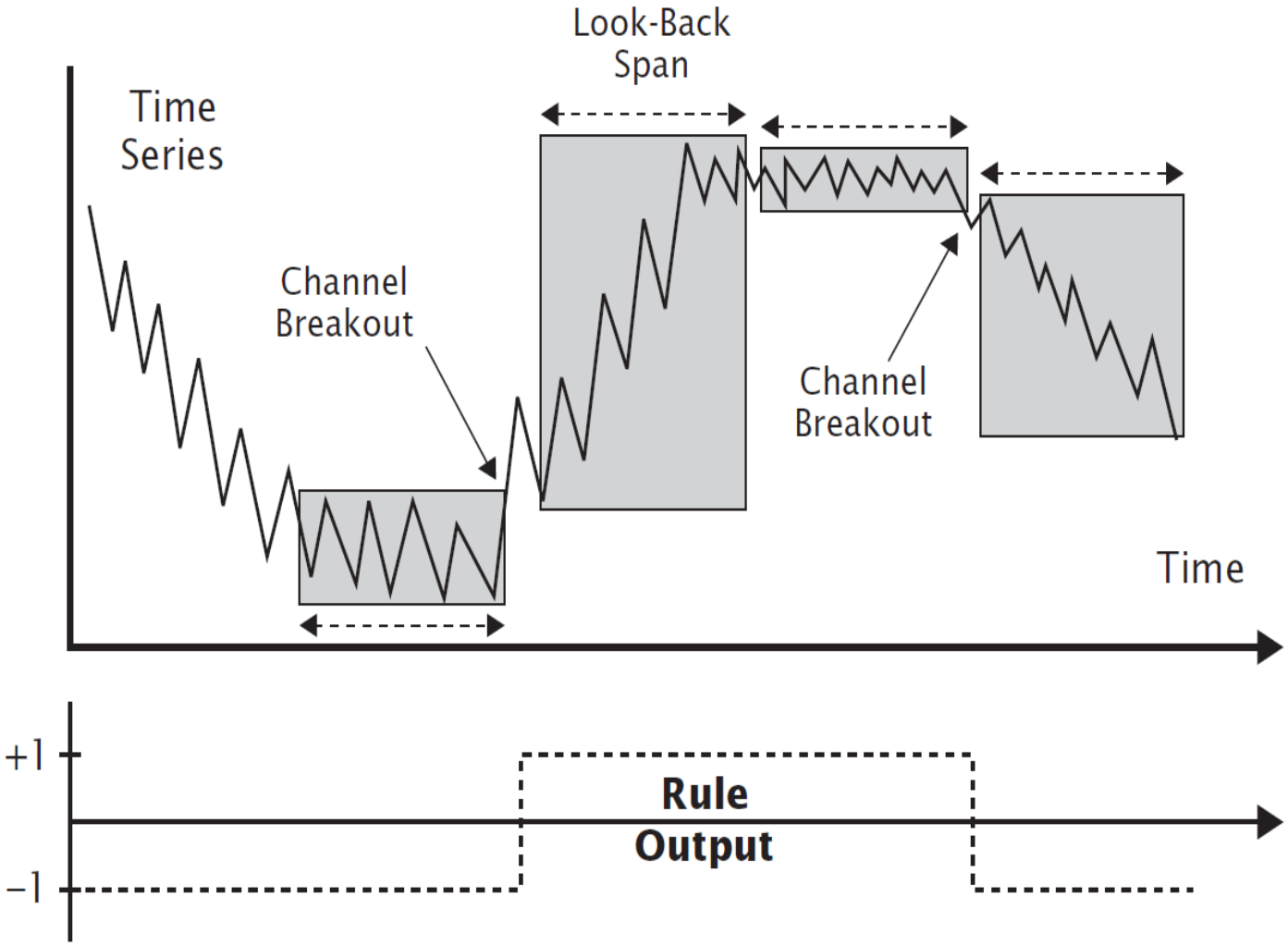


Figure: Channel-breakout operator

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THANK YOU